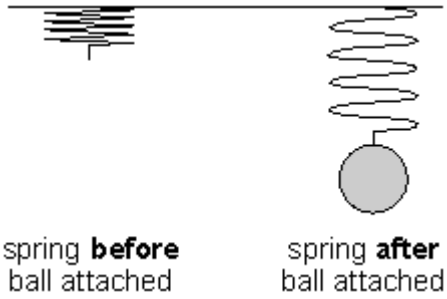
Q1. (a) John attaches a ball to a spring. The diagram below shows what happens.



(i) Which arrow shows the direction of the **force of the ball on the spring**?
Tick the correct box.



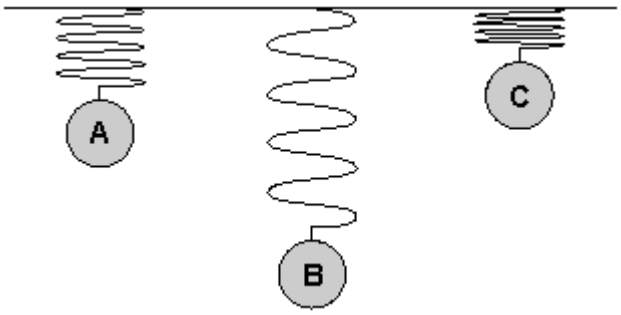
1 mark

(ii) Which arrow shows the direction of the **force of the spring on the ball**?
Tick the correct box.



1 mark

(b) The diagram below shows three metal balls attached to **identical** springs.



Which ball is the heaviest?
Write the letter.

.....

1 mark

Explain your answer.

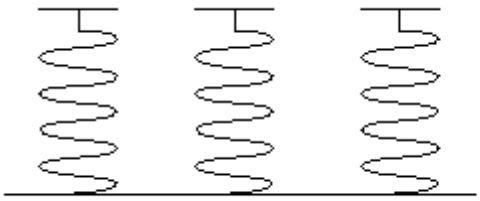
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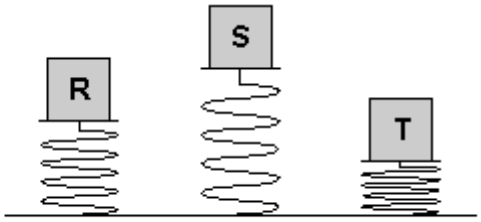
1 mark

(c) John has another three **identical** springs.
He puts a cube on each spring. Each cube has a different mass.

The diagrams below show the springs before and after John added the cubes.



springs **before** adding the cubes



springs **after** adding the cubes

Which cube is the heaviest?
Write the letter.

.....

1 mark

Explain your answer.

.....

.....

1 mark
maximum 6 marks

Mark schemes

Q1.

(a) (i) • ↓ ✓

1 (L3)

if more than one box is ticked, award no mark

(ii) • ↑ ✓

if more than one box is ticked, award no mark

1 (L3)

(b) • B

*accept 'the middle **or** second one'*

1 (L3)

any **one** from

- it hangs down the most
  accept 'the spring is wider apart'
- it stretches the spring most
  accept 'it stretches more'
- the spring is longer
  accept 'it stretched the longest'
  this mark cannot be awarded if the response conflicts with the first part of (b)
  a comparative answer is required for the mark
  non-comparative answers such as 'it is long' are insufficient

1 (L3)

(c) • T

accept 'the last one'

1 (L3)

any **one** from

- the spring was pushed down the most amount
  accept 'it squashes more'
- the cube pushed it down the most
  accept 'the spring is the tightest'
- the spring is shorter
  accept 'it is shortest'
  this mark cannot be awarded if the response conflicts with the first part of (c)
  a comparative answer is required for the mark
  non-comparative answers such as 'it is short' are insufficient

1 (L4)

[6]